



Marine Biology

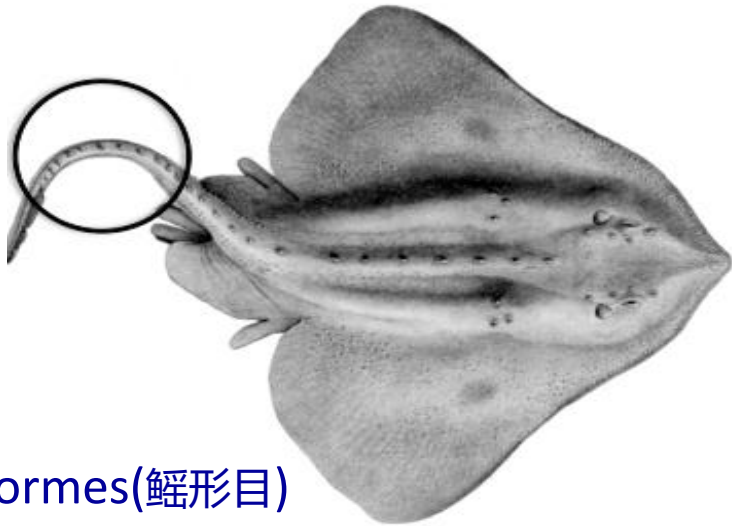
海洋生物学



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Rajiformes(鰐形目)

- Most have enlarged thorns along the midline of the back extending onto the tail
- Pelvic fins have two lobes (bilobate)
- Tail relatively stocky without a stinging barb, and usually with two small dorsal fins near its tip. The caudal fin is tiny, when present
- Mature male have enlarged spines near the eyes and pectoral wingtips
- **Lay eggs** (oviparous reproduction)



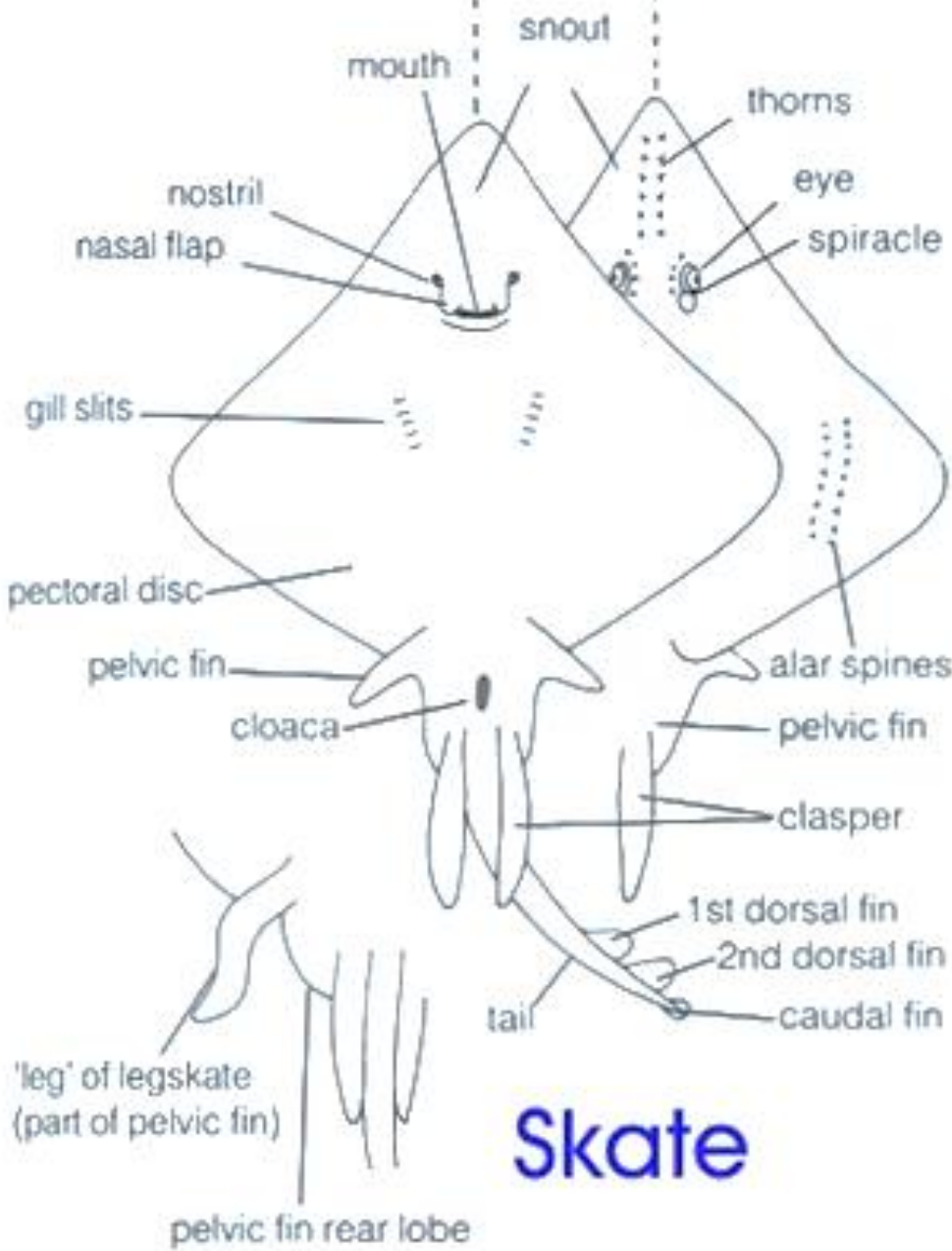
Myliobatiformes (鰐形目)

- No thorns along the midline of the back
- Each pelvic fin has only one lobe
- Tails is very slender and whip-like with a stinging spine midway along its length and usually without a dorsal fin. When a dorsal fin occurs, it is near the base of the tail. The caudal fin is either reduced and contiguous or absent
- Mature males do not have malar (颊) or alar (翼) spines
- **Give live birth** (viviparous reproduction)

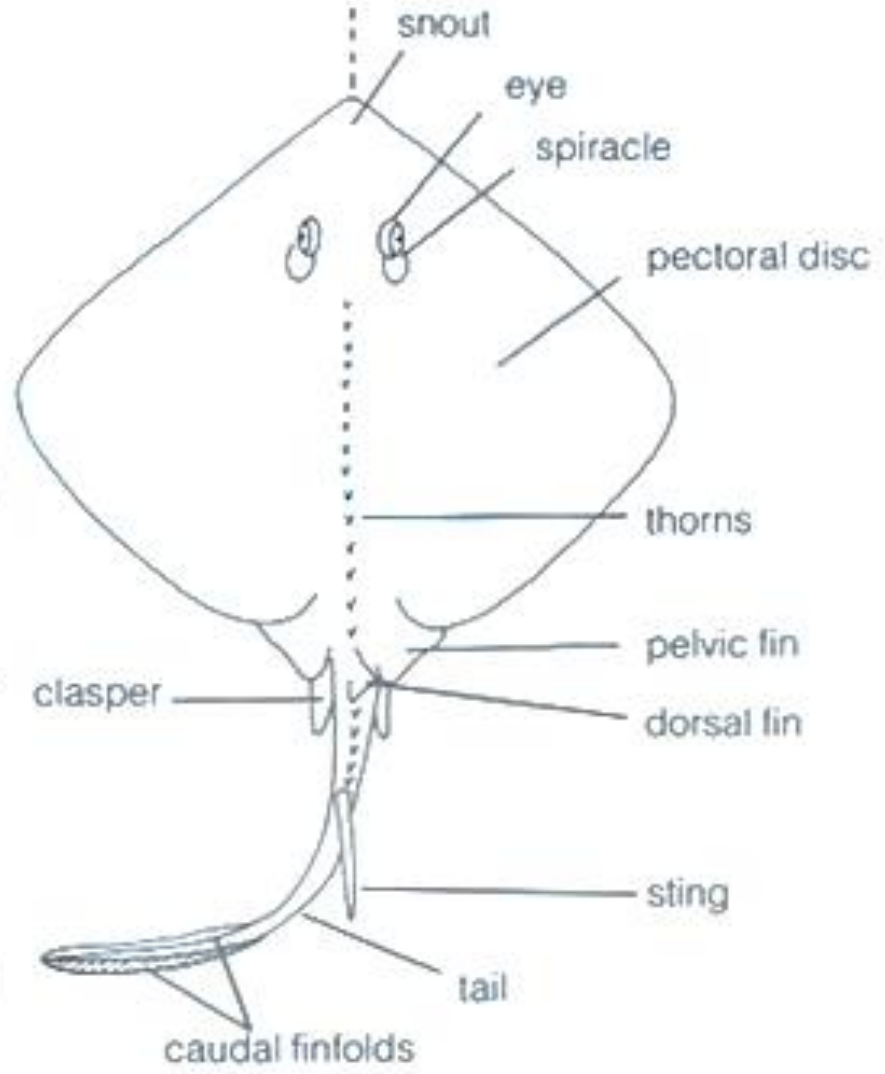
VENTRAL VIEW

DORSAL VIEW

DORSAL VIEW



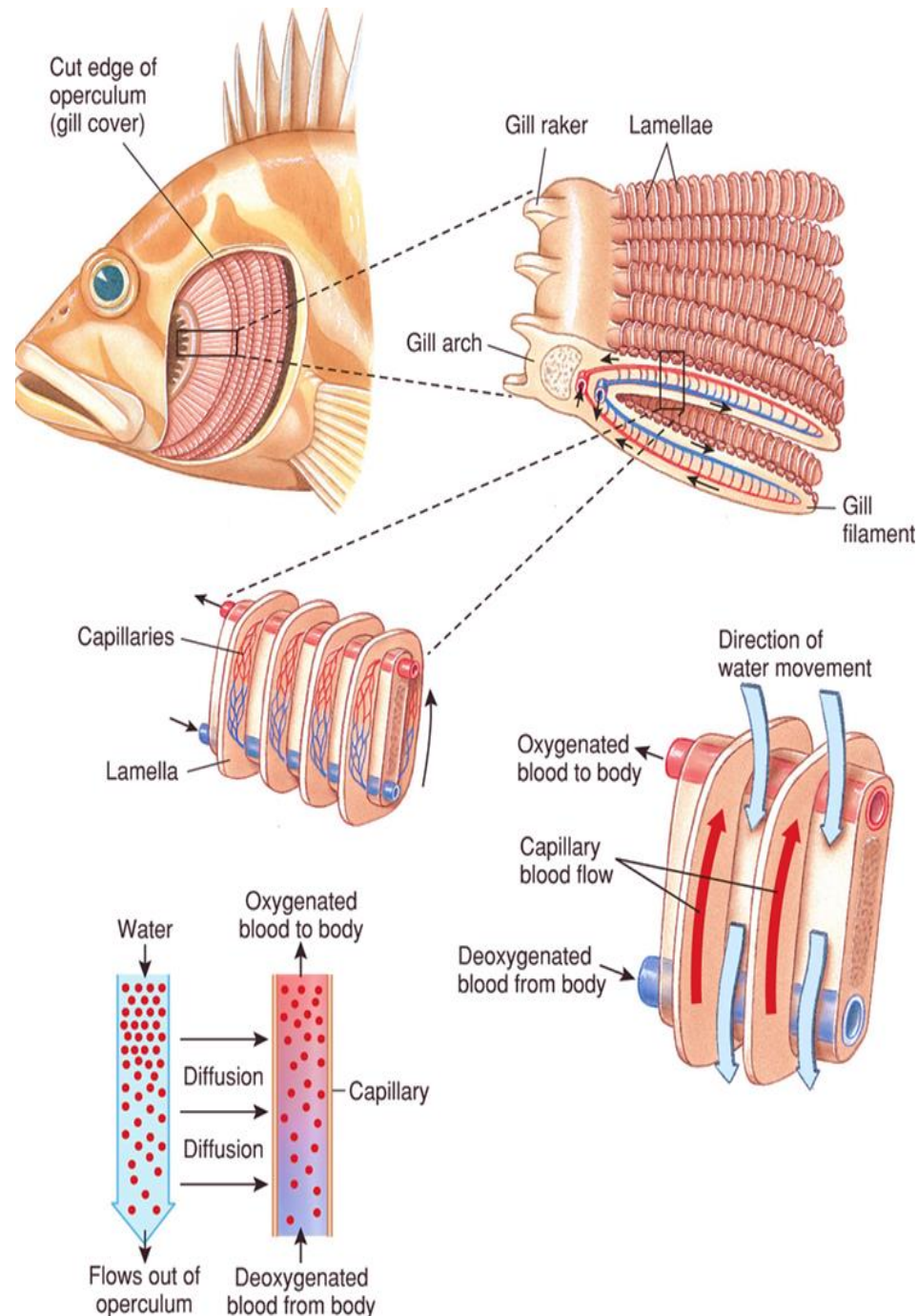
Skate



Ray

6. Fish Gills

- The construction of the gill is the same in all fish
- **Gill arch (鳃弓)** supports the entire structure, **gill rakers (鳃耙)** are on the forward surface of the gill arch and **gill filaments (鳃丝)** trail behind the gill arch
- Exchange of oxygen and carbon dioxide takes place on these surfaces
- **Count-current flow system**



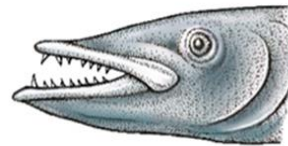
7. Acquiring and Processing Food-Digestion system

- **Mouth structure** reveals the dietary preferences of fish

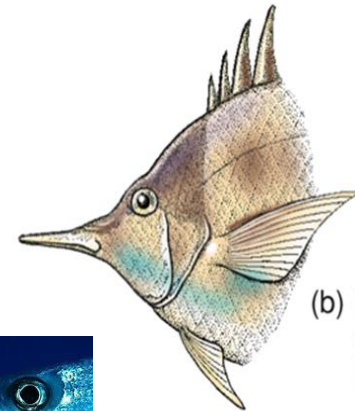
- The barracuda (梭鱼) uses rows of sharp teeth and a wide mouth to capture its prey – other fish

- The butterfly fish uses its long tube-like mouth to feed on corals to feed on corals

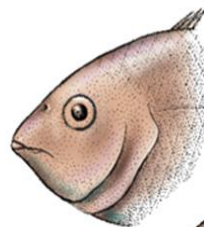
- The “beak” (fused teeth) seen in parrotfish (鹦嘴鱼) allows for these fish to scrape algae and other organisms off of hard surfaces



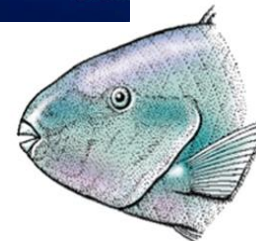
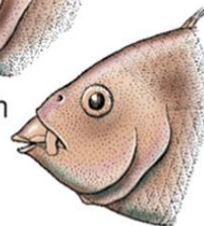
(a) Barracuda



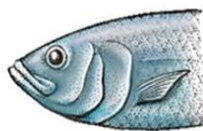
(b) Long-billed, six-spined butterflyfish



(c) Slipmouth



(d) Parrotfish



(e) Herring

- The **position of the mouth** is also important

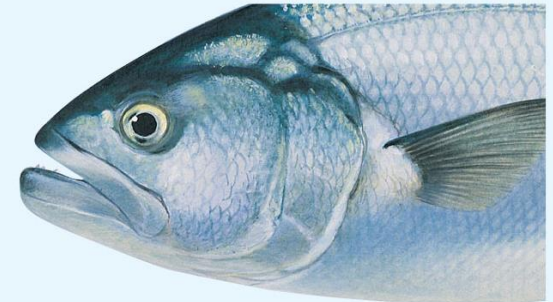
- ▣ A strongly forward facing mouth is important in fish who chase down their prey (as seen in barracuda)

- ▣ A downward facing mouth would be seen in fish feeding at/near the bottom

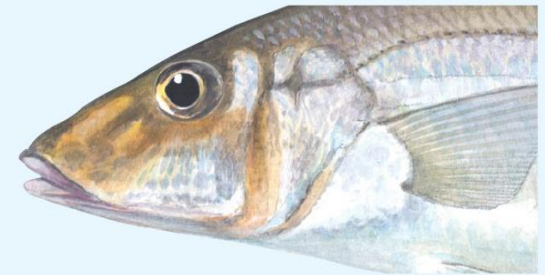
Superior – the mouth is positioned towards the surface and the fish feed on what is above them, e.g., garfish and snook.



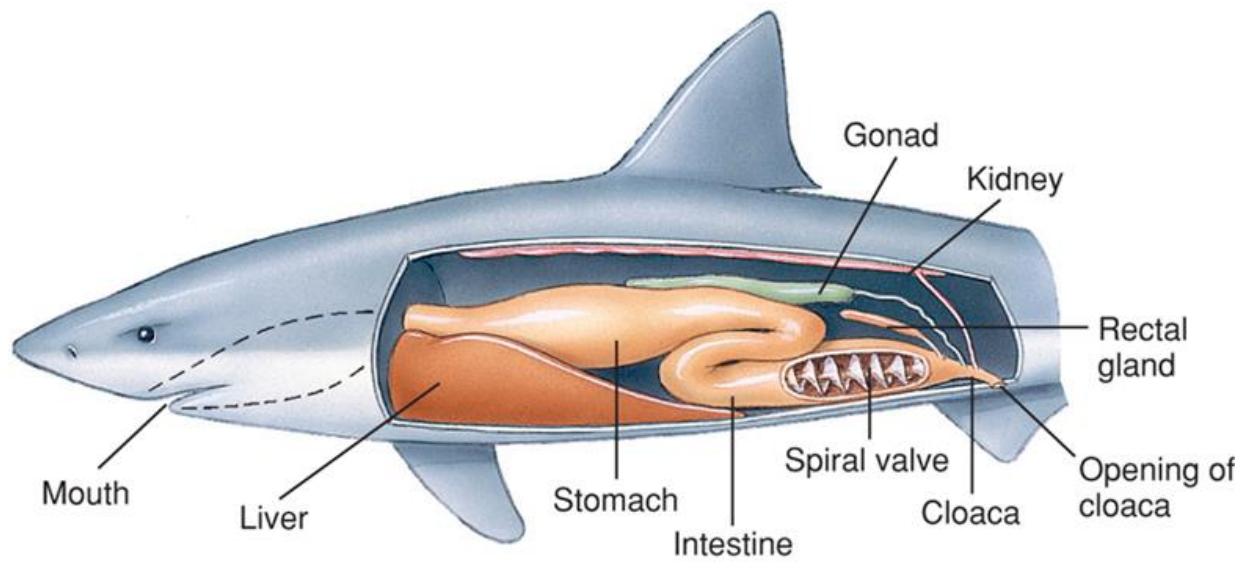
Terminal – the mouth is positioned in the middle of the head and the fish either chase prey or feed on what is ahead of them, e.g., tuna and tailor.



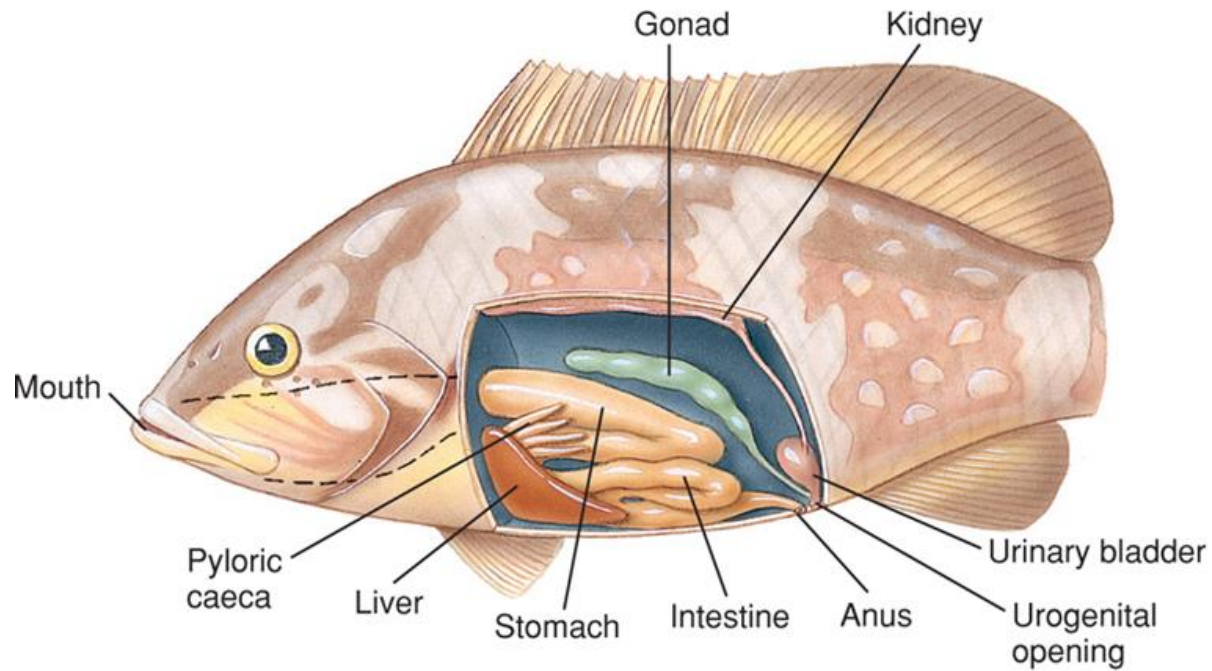
Inferior – the mouth is positioned towards the bottom and the fish prey upon or scavenge/graze benthic food sources, e.g., whiting and cobbler.



- Digestion of food in fish is completed with the aid of a stomach, intestine (with anus), liver, pyloric caeca (幽门盲囊) and pancreas (胰腺)
- **The stomach is structured very similarly to human** : stretch muscle in the wall of the stomach; the churning motion of the stomach wall will mechanically digest presented food
- The intestine, pyloric caeca, pancreas and liver all **secrete digestive enzymes** to aid in the digestion process
- The intestines of **carnivorous fish** tend to be short and straight (normally assisted by **spiral valves** for digestion), while the intestines of **herbivorous fish** are longer and more coiled (plant and algae material is more difficult to process, so it needs to stay in the intestines longer)



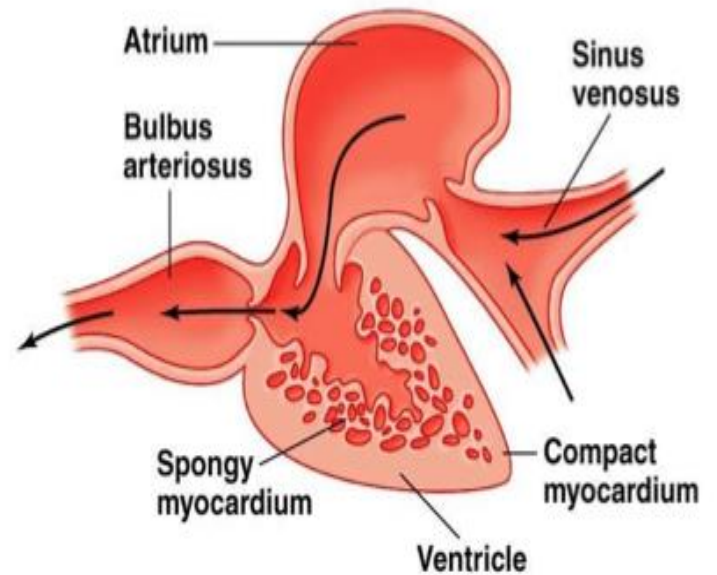
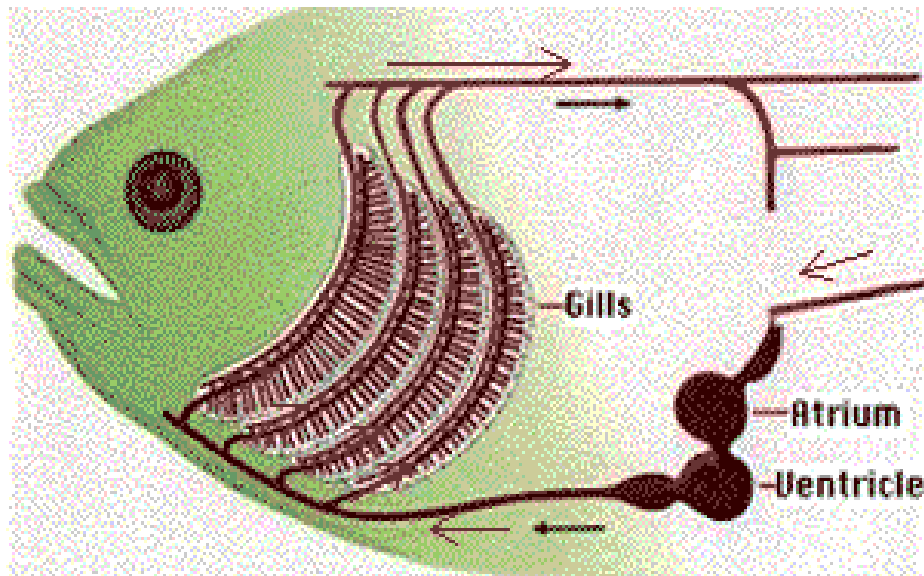
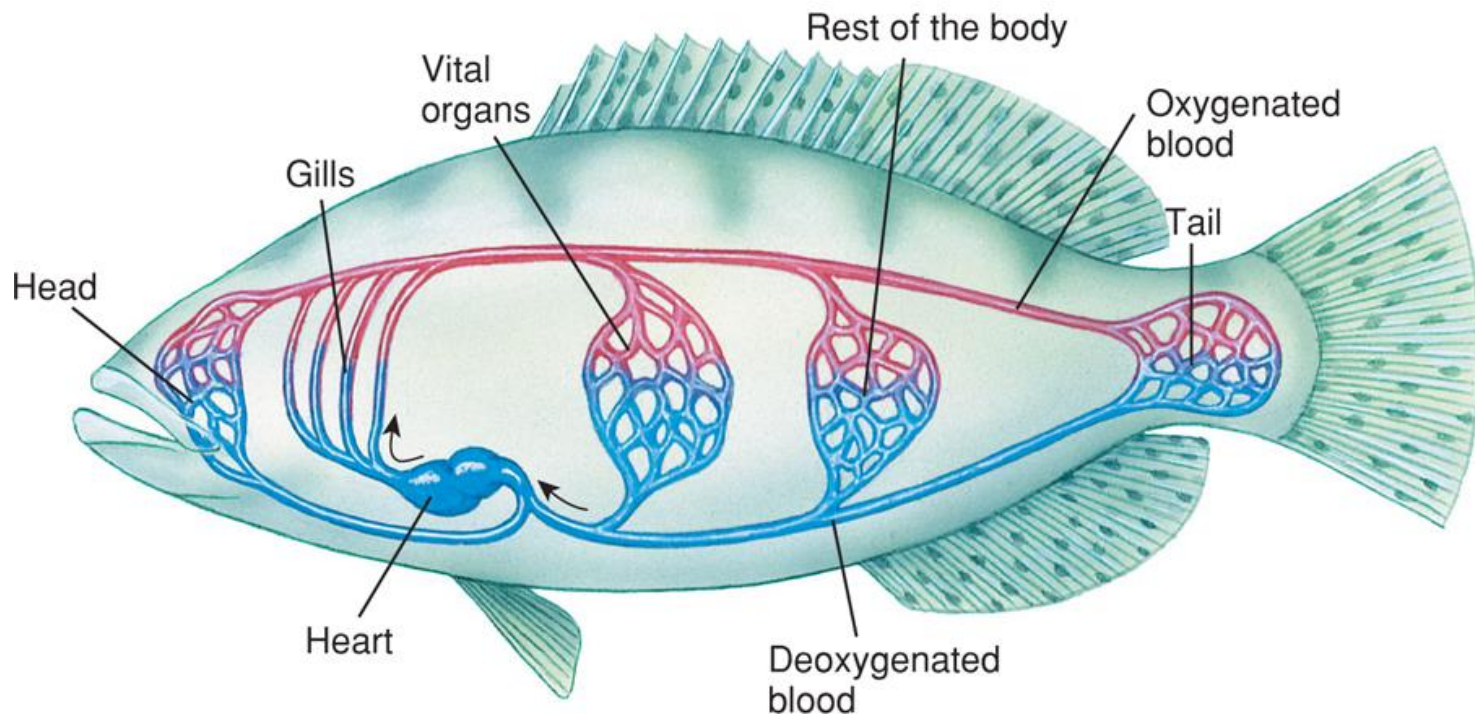
Cartilaginous fishes



Bony fishes

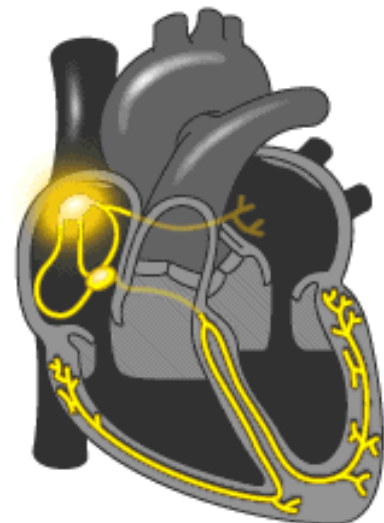
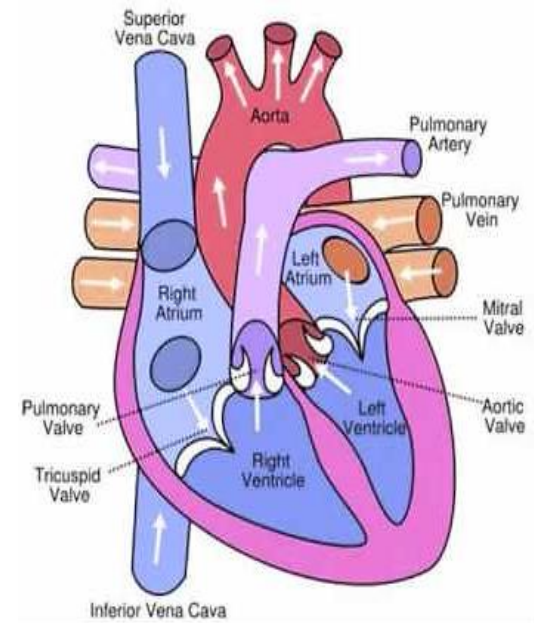
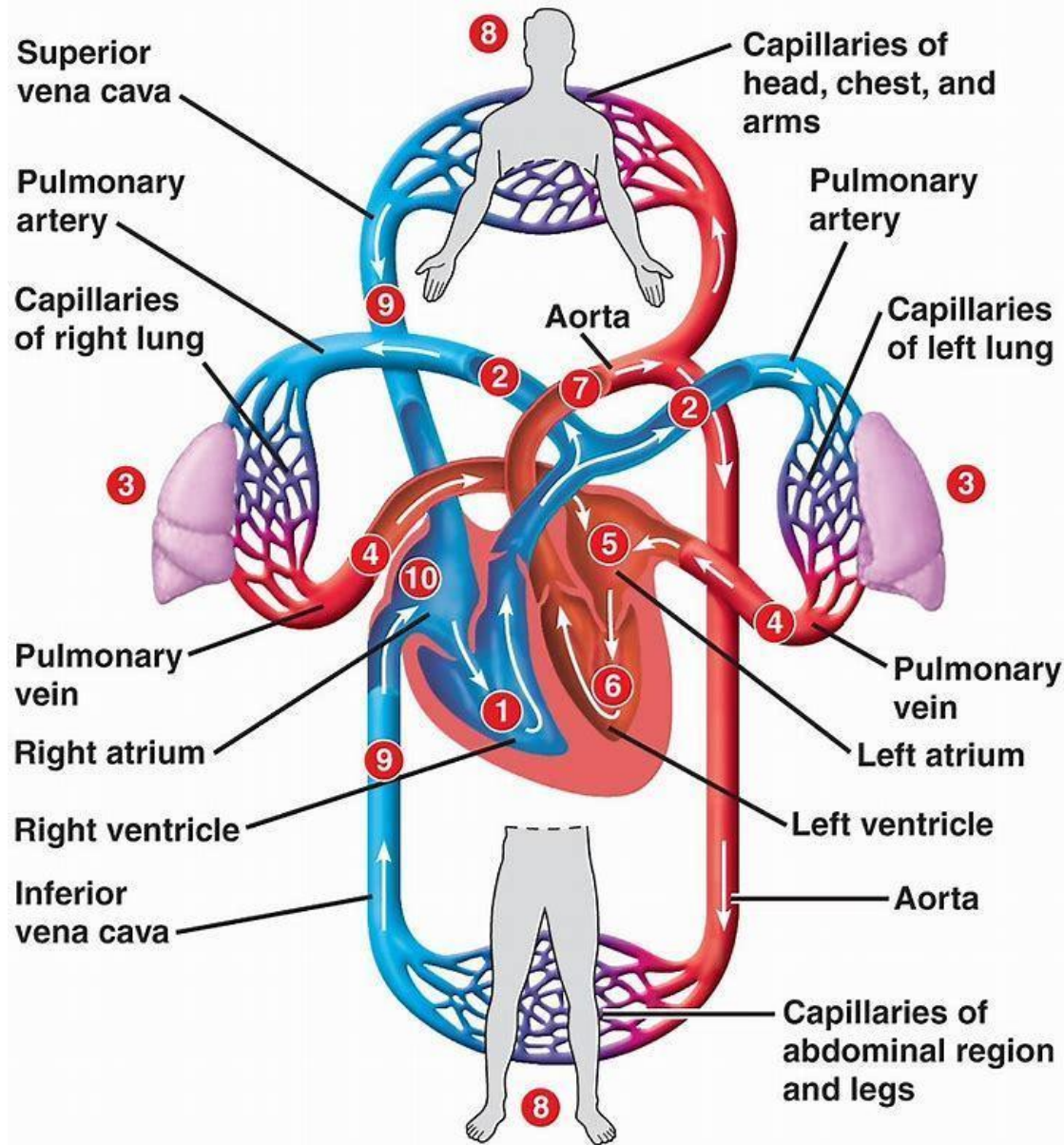
8. The Circulatory System

- Fish have a **two chambered heart** that serves to pump blood throughout the body (in contrast to the 4 chambered heart seen in mammals)
- A system of arteries, veins and capillaries takes blood to the body tissues and returns it for re-oxygenation by the gill filaments
- Oxygen and carbon dioxide will diffuse across the thin membranes of the capillaries either in the gills or at the tissues of the body

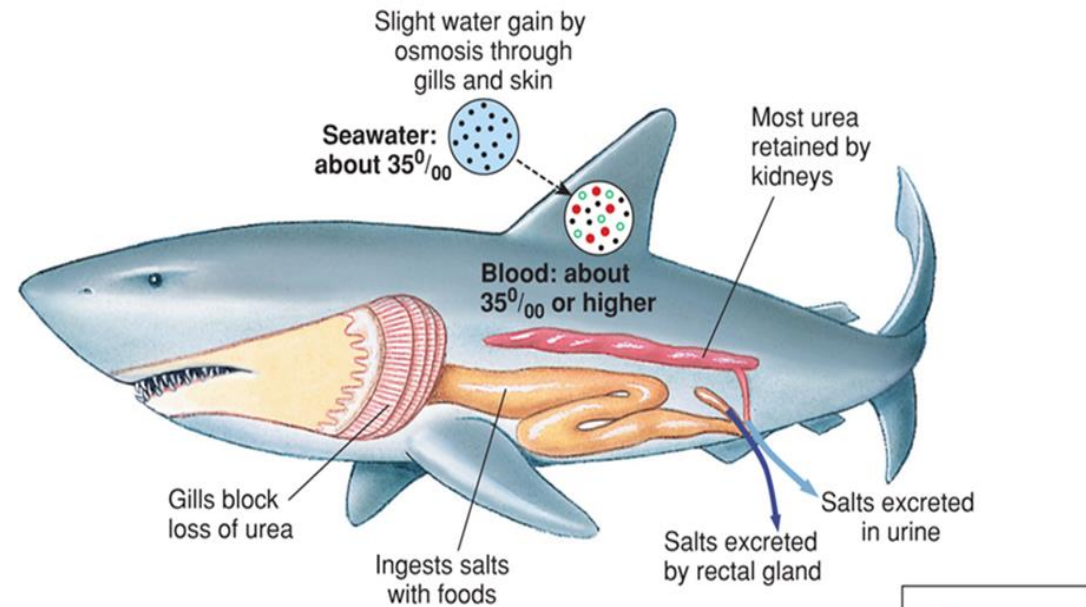


(a) Bony fish heart

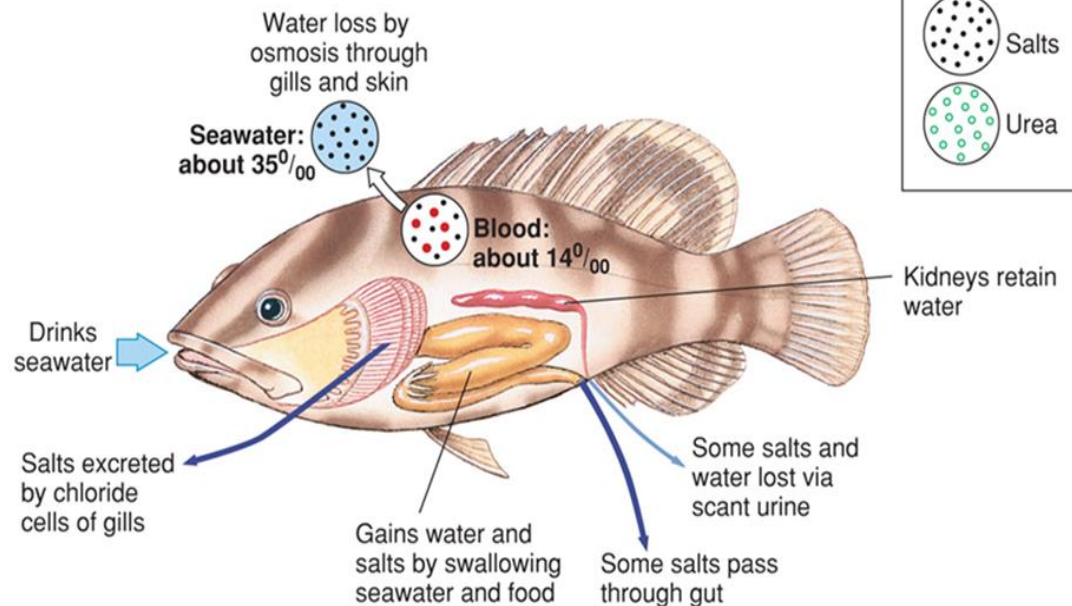
- Double circulatory system



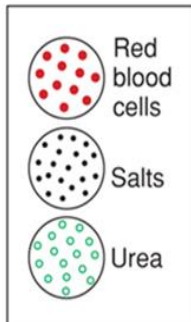
- Fluid balance in fish
- ▣ The forces of osmosis and diffusion are always at work in an organism
- ▣ Since marine fish are living in an environment where the water is very solute-rich, these fish have a tendency to gain solutes and lose water



(a) **Cartilaginous fishes**

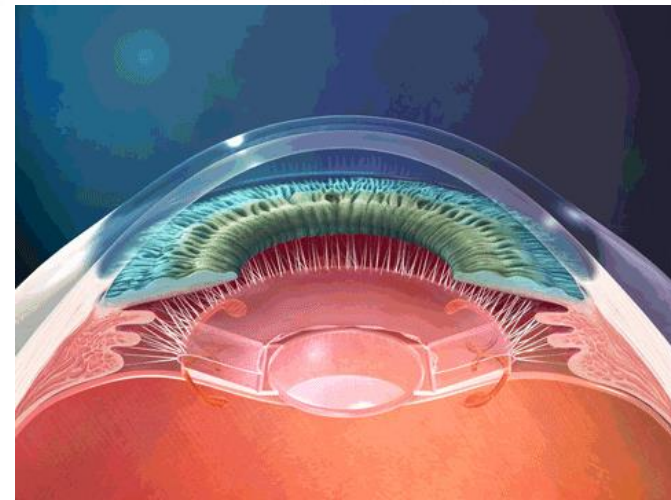
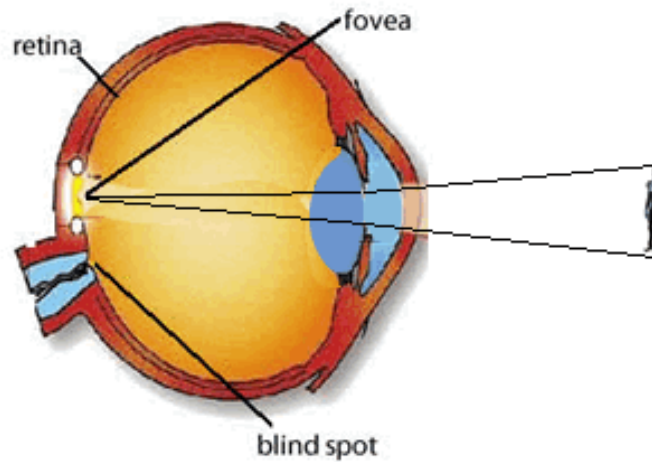
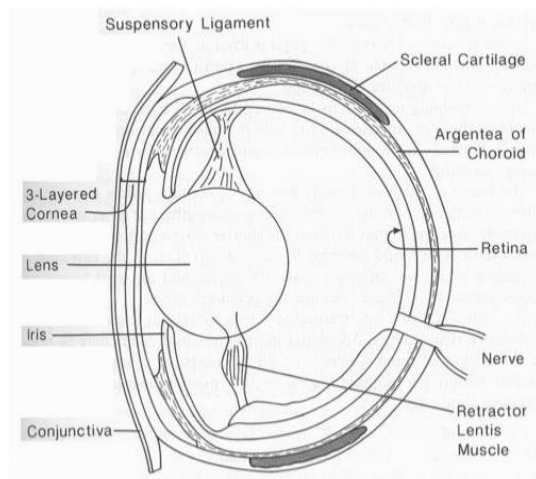
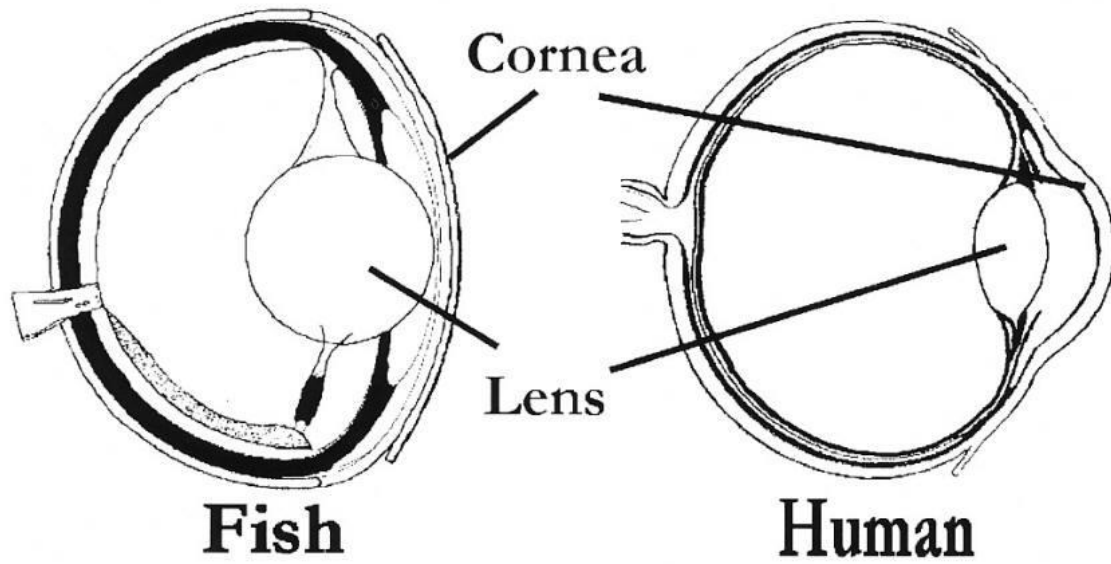


(b) **Bony fishes**



9. The Fish Nervous System

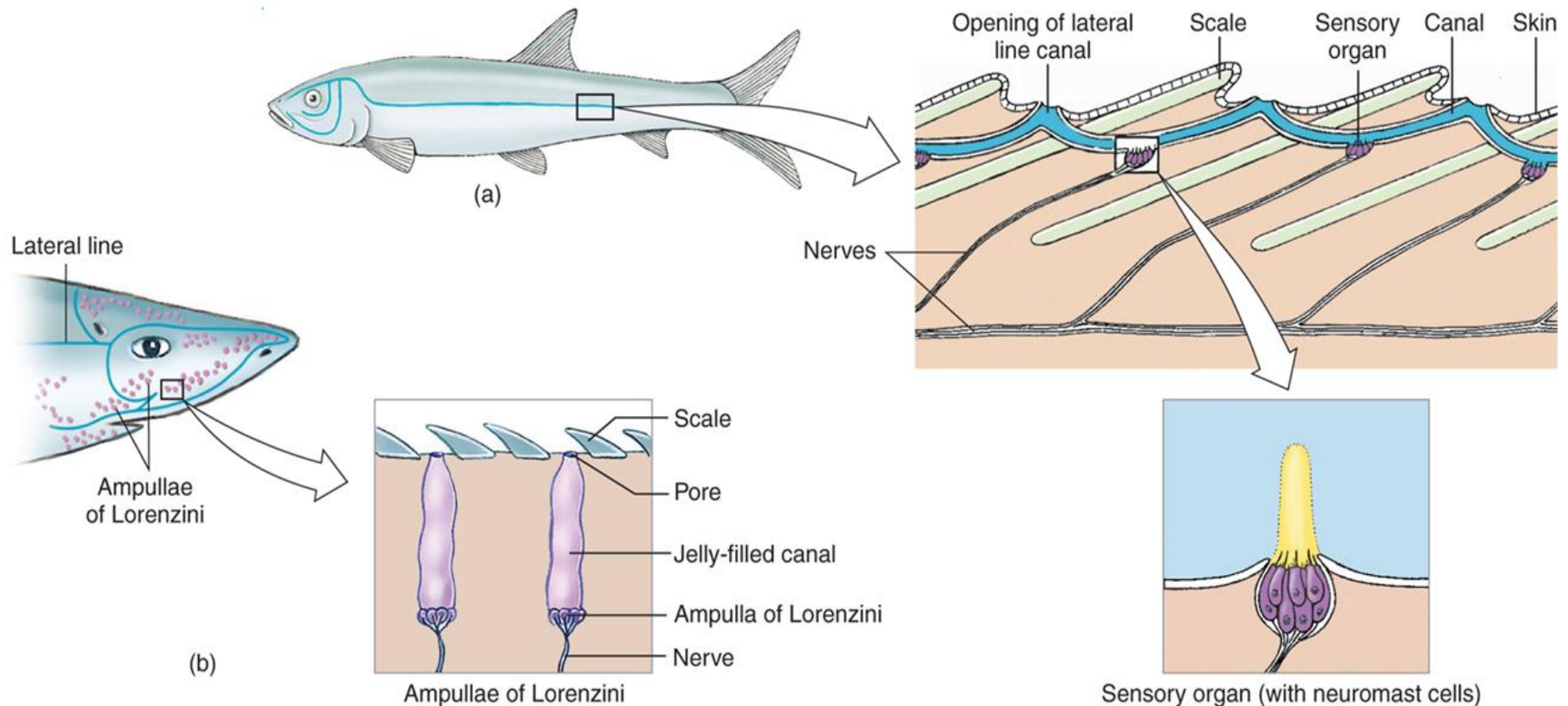
- The fish have a **brain, spinal cord (脊髓)** and **numerous nerves** like other vertebrates
- Fish also possess **olfactory (嗅觉的) sacs** (with nostrils, **鼻孔**) for smelling
- **Taste buds** are located in the mouth, lips, barbels (**触须**) and skin
- The **eyes** are structured slightly differently
 - ❑ In humans, the lens changes shape for focusing on items
 - ❑ In fish, the position of the lens changes like in a camera



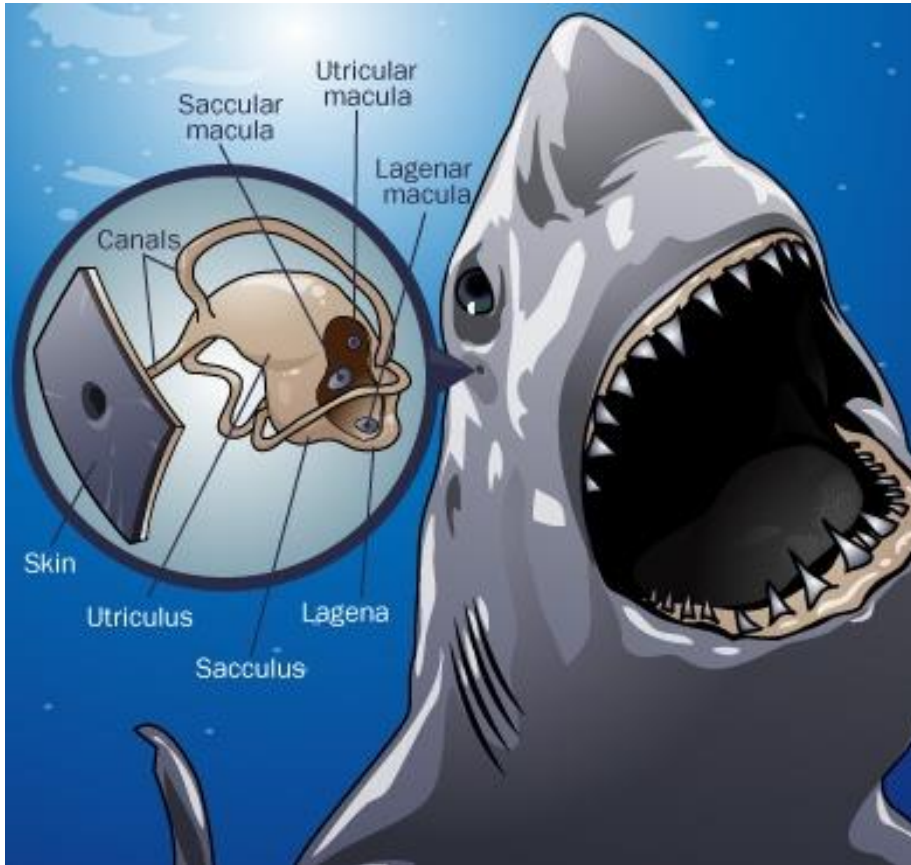
- In some sharks, the eye is covered by a **nictitating membrane** (瞬膜) that covers the eye – this clear membrane helps these predators protect the eye, especially during feeding. Incidentally, this membrane is also seen in some reptiles



- All fish rely heavily on the **lateral line system (侧线系统)**:
The lateral line is a series of pores and canals lined with cells called **neuromasts (神经丘)** that are specialized to detect vibrations, which can indicate a predator or prey or the position of other fish in a school (**鱼群**)



- The **inner ears** are set in fluid-filled canals with sensory cells similar to the lateral line system

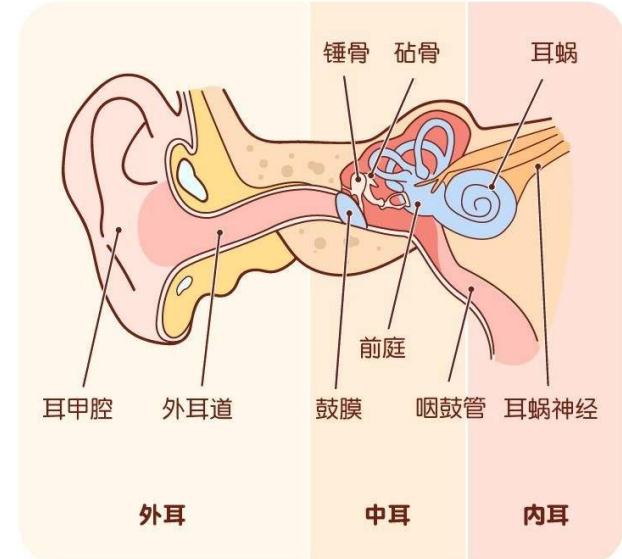
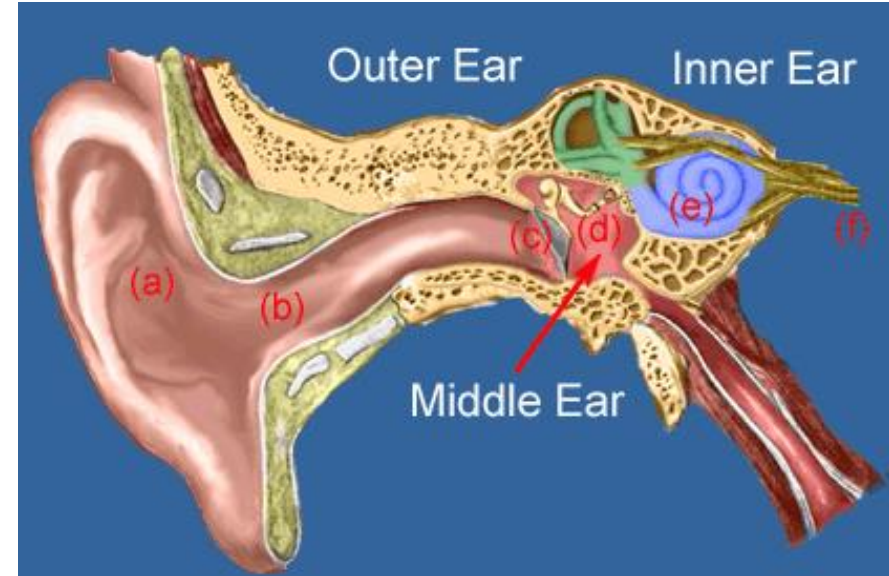


Utriculus-椭圆囊

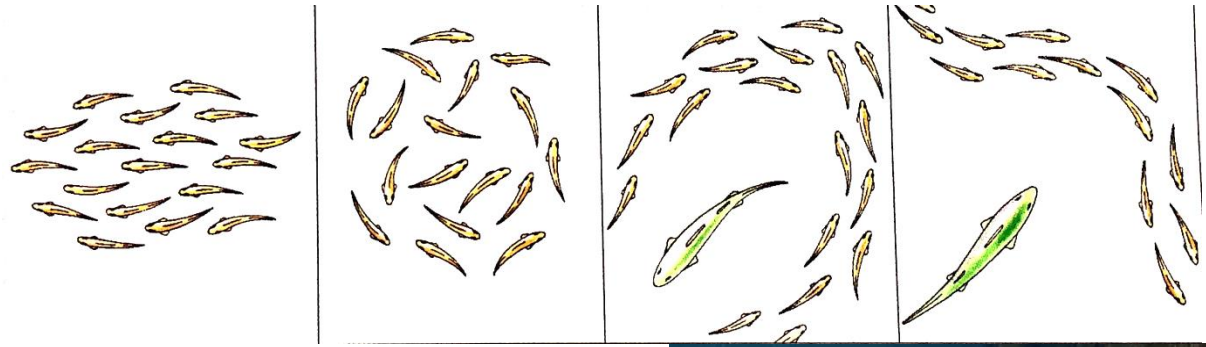
Sacculus-球状囊

Legena-瓶状囊

Macula-斑点



10. Behaviors in Fish



1) Schooling

- Schooling is used by a wide variety of fish
- Schooling makes it possible for a group of smaller fish to appear much larger (as thus avoid detection by predators)
- It also makes it harder for a predator to capture any one fish
- Because of this, many fish school as juveniles
- About 4000 species school as adults



2) Territoriality (领地行为)

- Some fish are territorial by nature all the time, others are only territorial during reproduction
- Fish maintain their territories normally by “posturing” (raised fins, open mouth, darting, etc.) to show their aggression
- Fights between individuals are actually rare





(a)



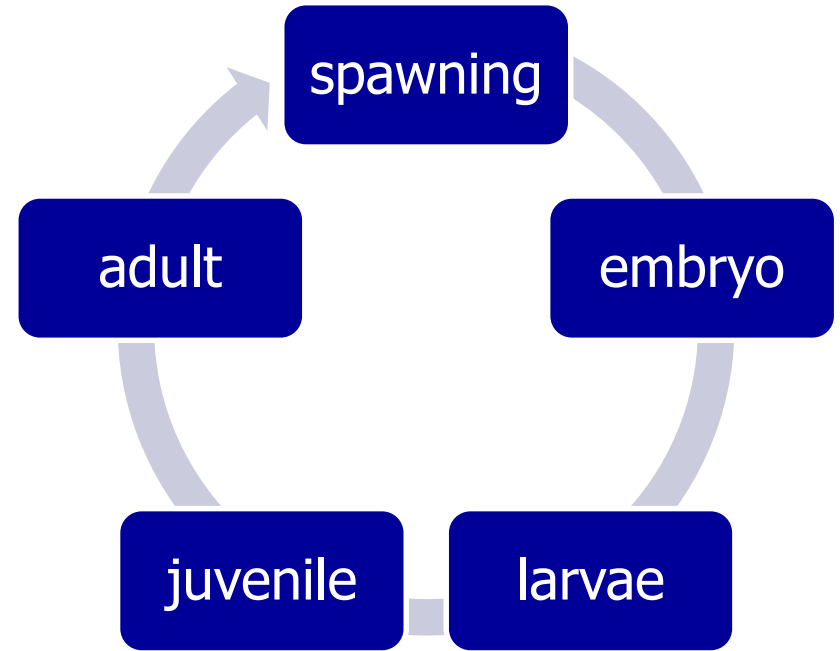
(b)

Adults of some butterflyfishes occur in male and female pairs. Experiments using transparent plastic cages provide valuable information on their behavior. Here, a caged oval butterflyfish elicits an aggressive response from an individual chevron (人字形) butterflyfish, an indication that the caged fish has intruded in the territory of the latter.

3) Migrations

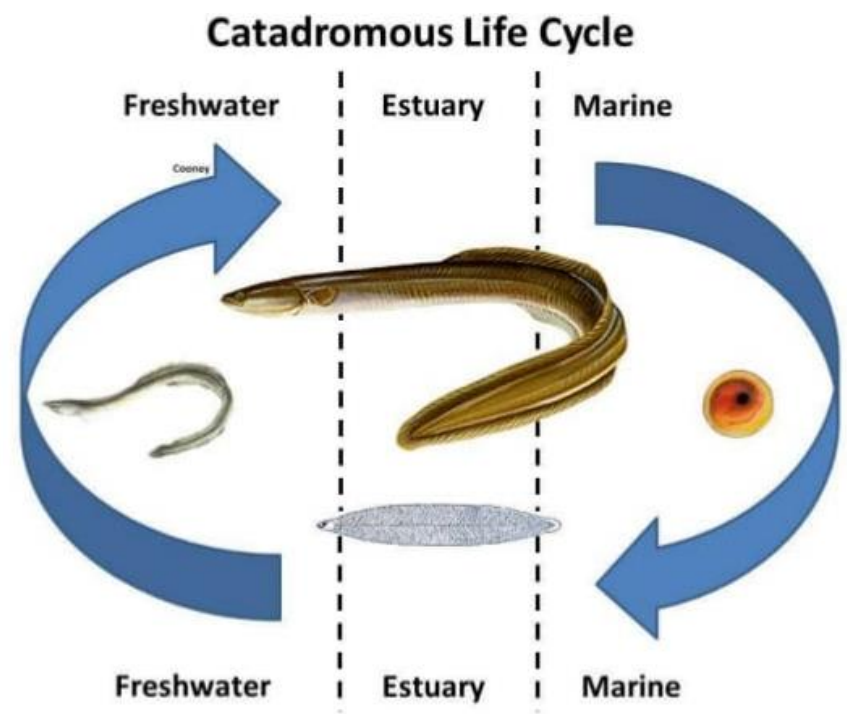
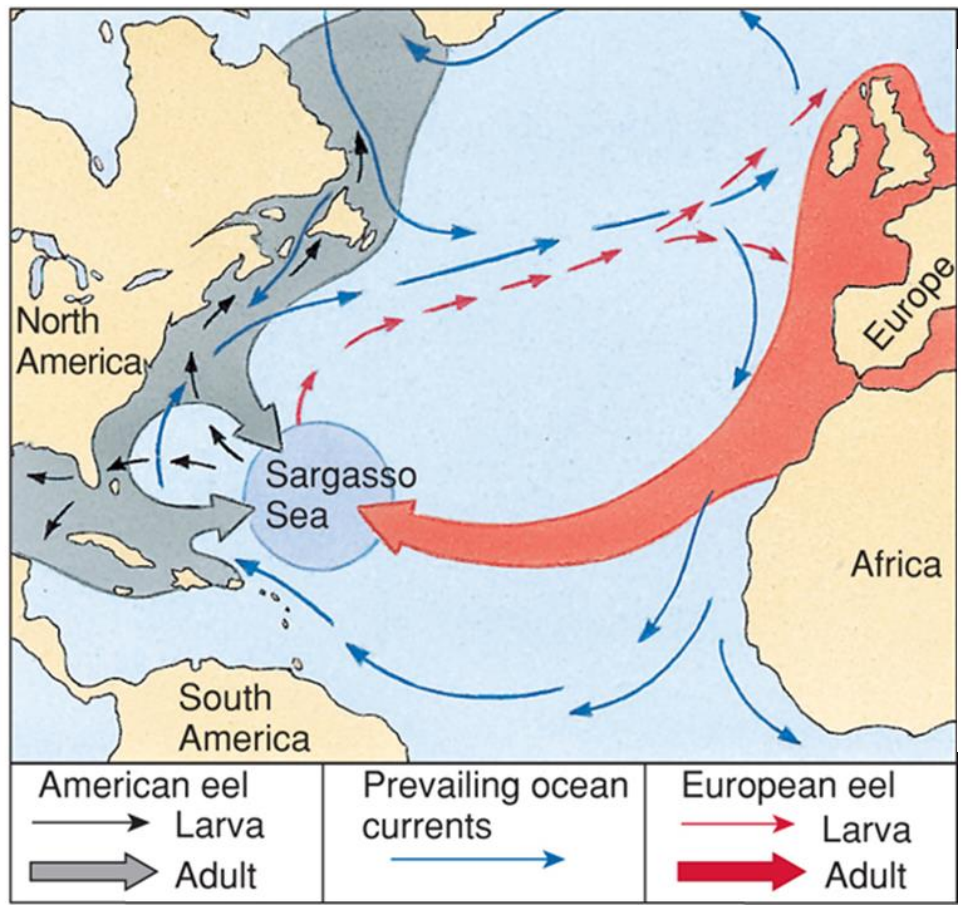
- Life in the cycle of **spawning-embryo-larvae-juvenile-adult**
- Some species migrate between freshwater and saltwater at different times in their life
- Reasons for migration are:

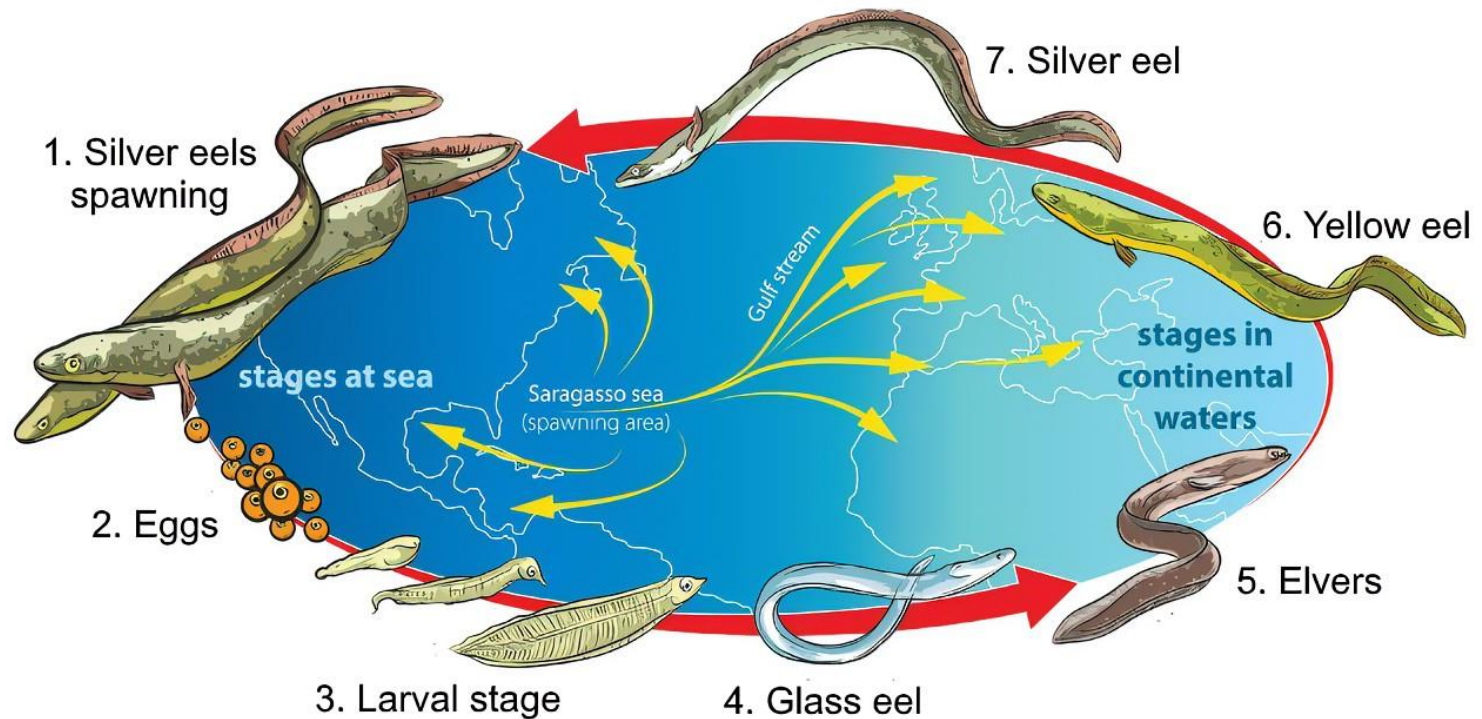
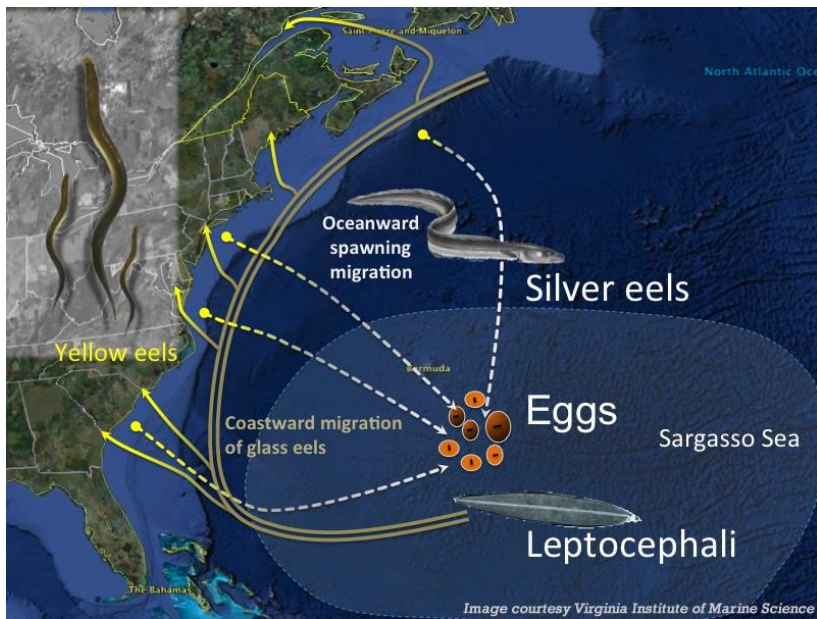
- ✓ To avoid unfavorable conditions: light, temperature, pH, water currents *etc.*
- ✓ Better breeding
- ✓ To enhance the survival of offspring's
- ✓ Scarcity of food in feeding areas
- ✓ Population pressures (competition and predation)



❑ **Catadromous species (降海性)** :fish are born in saltwater, then migrate into freshwater as juveniles where they grow into adults before migrating back into the ocean to spawn

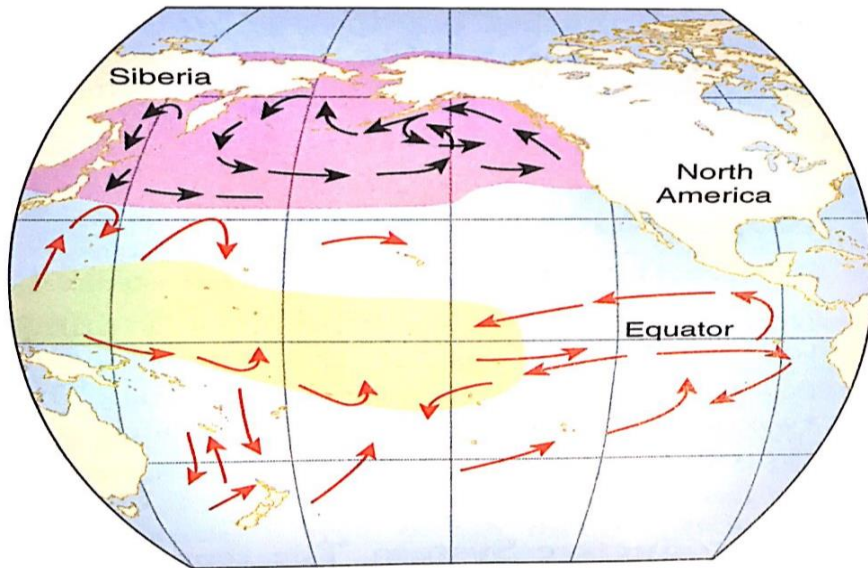
Example: *American eel*, *European eel*, *inanga*, *shortfin eel*, *longfin eel*





□ **Anadromous species (溯河性):** fish are born in freshwater, then migrate to the ocean as juveniles where they grow into adults before migrating back into freshwater to spawn

Example: *salmon, smelt, American shad, hickory shad, striped bass, lamprey, gulf sturgeon*

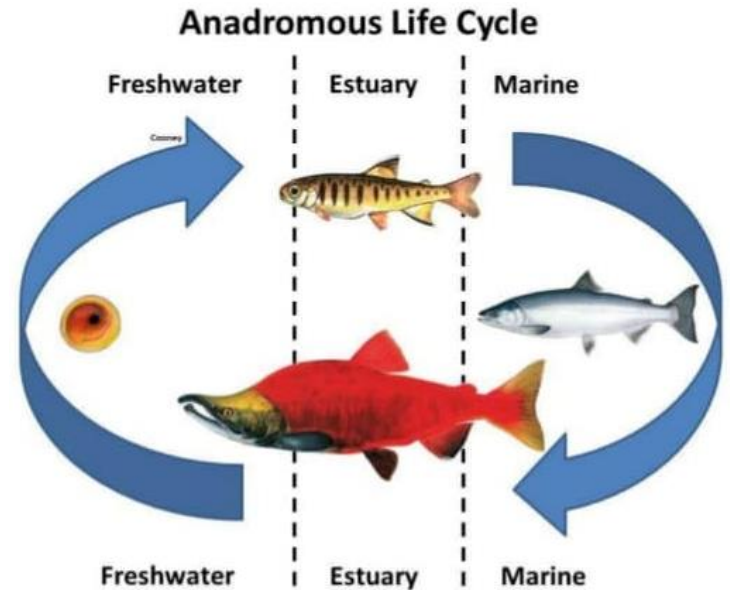


Pacific salmon

- Major migration routes
- Range

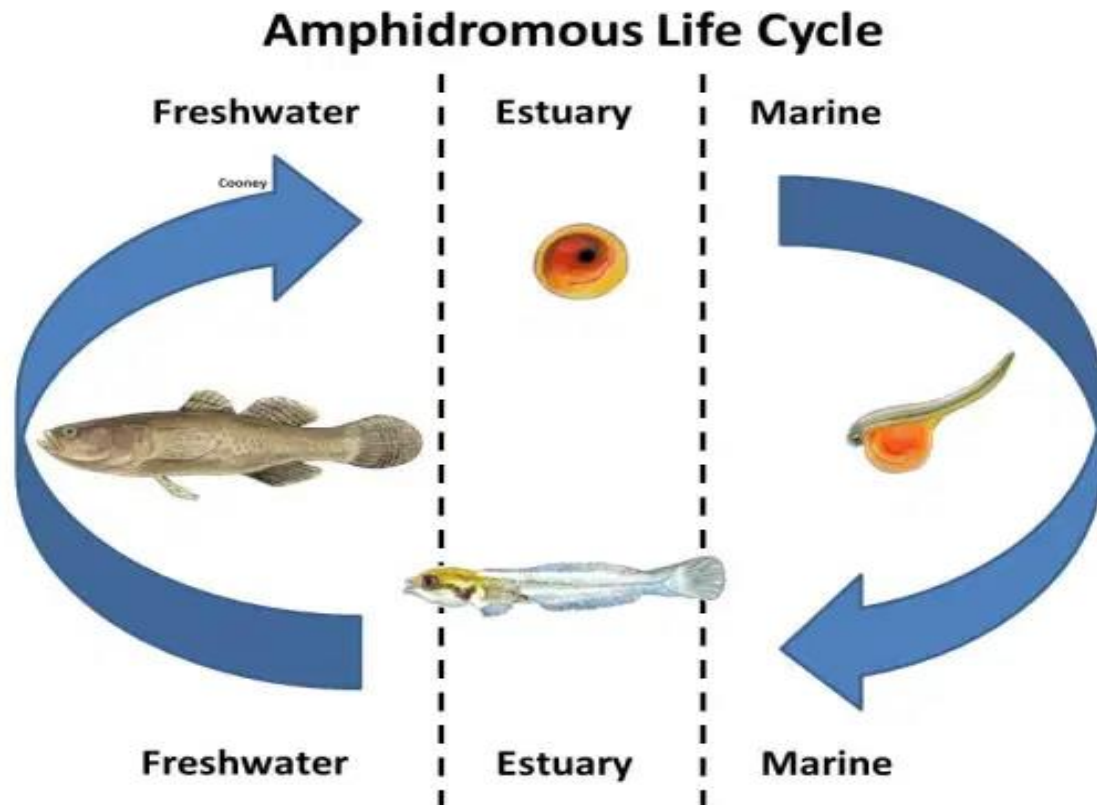
Skipjack tuna

- Spawning grounds
- Major migration routes



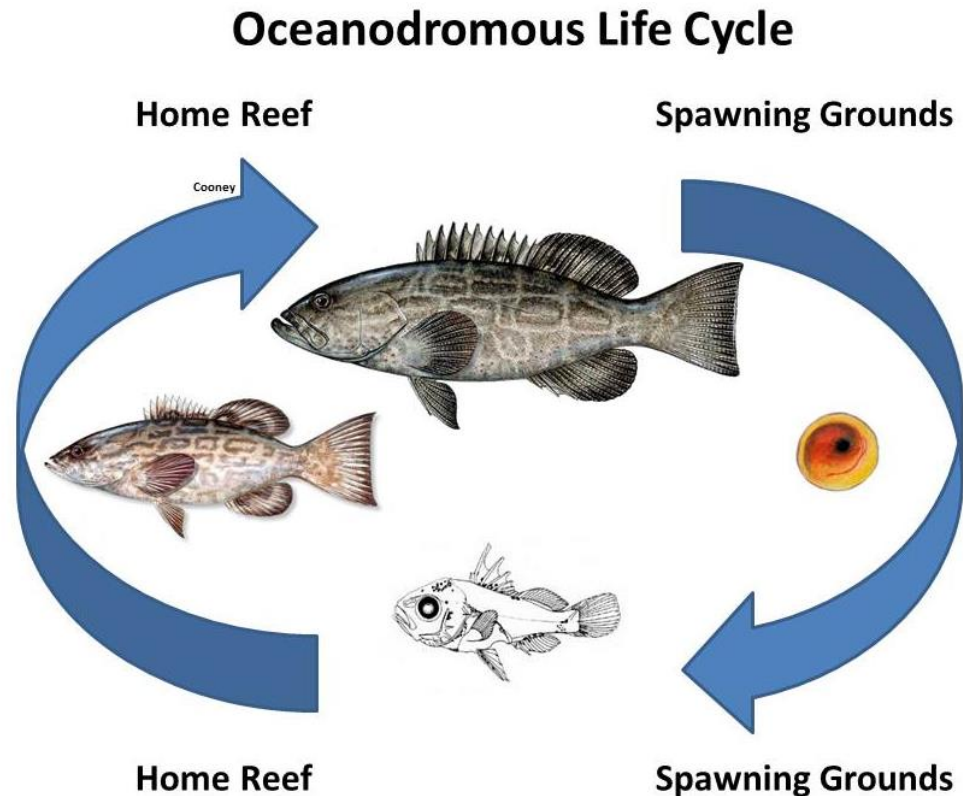
□ **Amphidromous species (两栖洄游)** :Fish are born in freshwater/estuaries, then drift into the ocean as **larvae** before migrating back into freshwater to grow into adults and spawn

Examples: *bigmouth sleeper*, *mountain mullet*, *sirajo goby*, *river goby*, *torrentfish*, *Dolly varden*



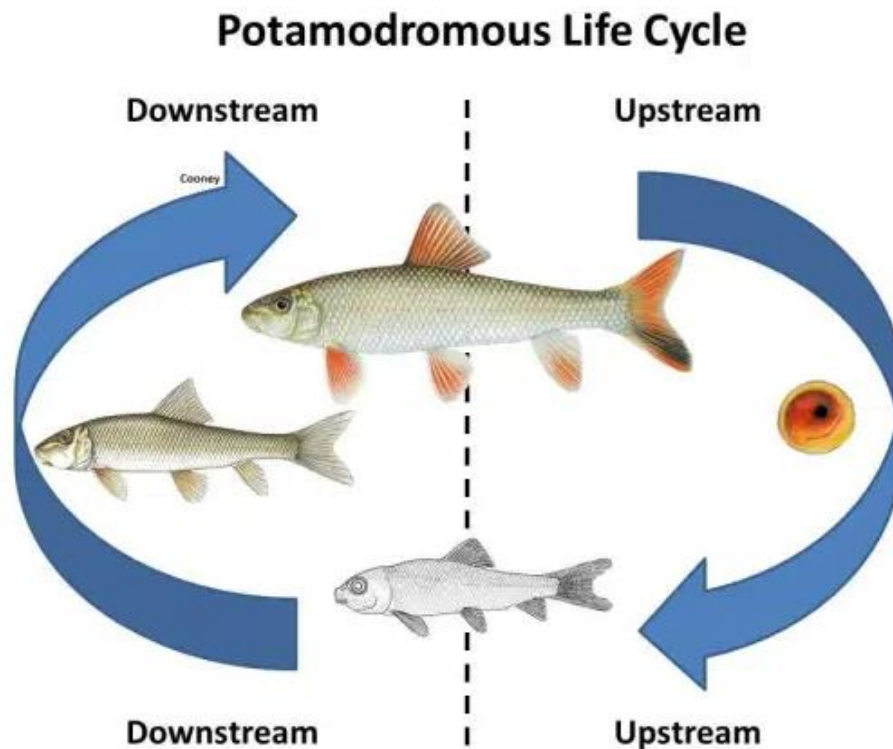
- ❑ **Oceanodromous species (seawater species, 海洋洄游):** Fish are born near spawning grounds, then drift on ocean currents as larvae before settling as juveniles to grow into adults before migrating back to spawning grounds

Examples: *black grouper*, *mutton snapper*, *goliath grouper*

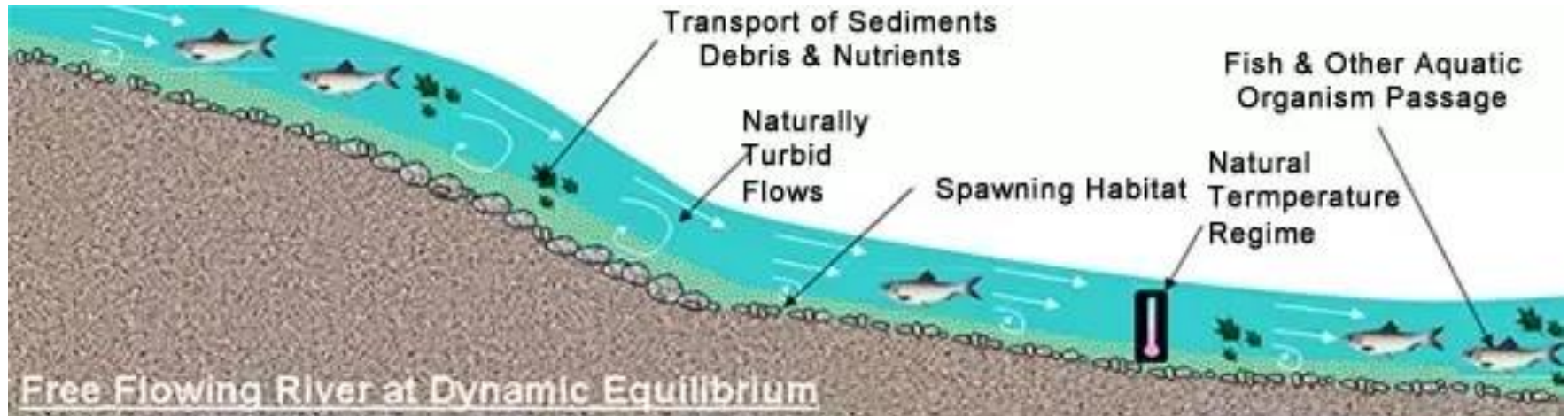


□ **Potamodromous species (freshwater species, 淡水洄游):** Fish are born in upstream freshwater habitats, then migrate downstream (still in freshwater) as juveniles to grow into adults before migrating back upstream to spawn

Examples: *sicklefin redbhorse* (吸口鱼), *lake sturgeon* (鲟鱼), *robust redbhorse*, *flathead catfish*



Human impact to fish migrations



UPSTREAM IMPACTS*

Reduced:
Natural Function, Water Quality, Oxygen, Turbid Flow, Circulation, Available Habitat
Rivers ability to adjust horizontally and vertically (reduced resilience to change)

Increased:
Pollutant Accumulation, Stratification, Temperatures, Algae Blooms

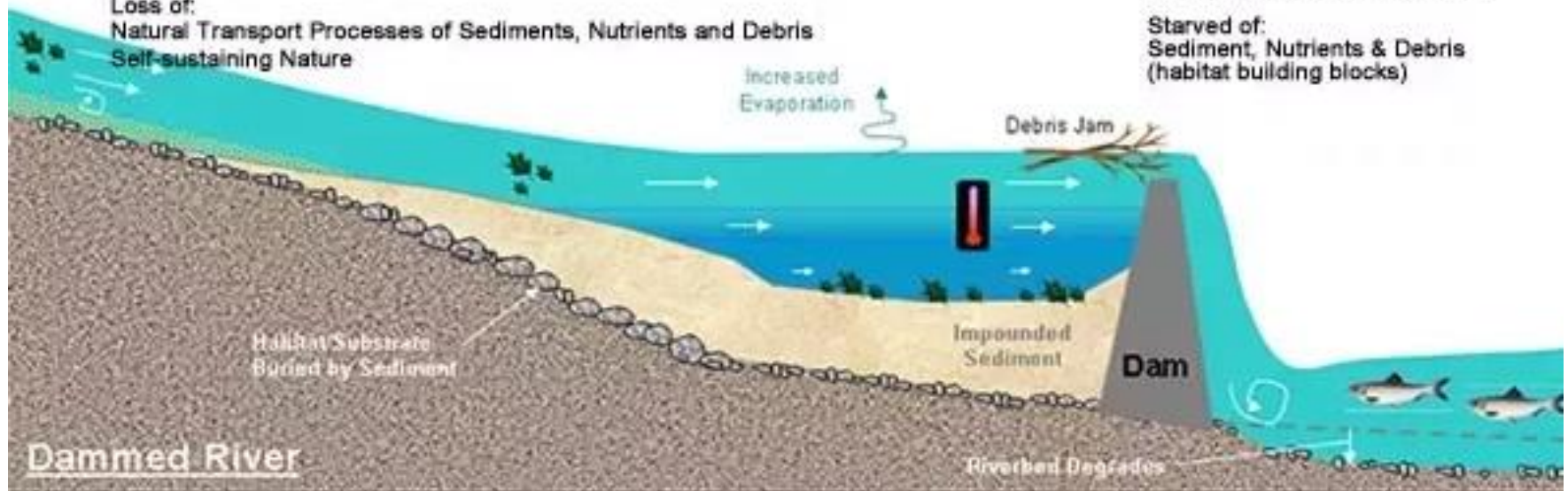
Loss of:
Natural Transport Processes of Sediments, Nutrients and Debris
Self-sustaining Nature

DOWNSTREAM IMPACTS*

Reduced:
Water Quality & Riverbed Elevation

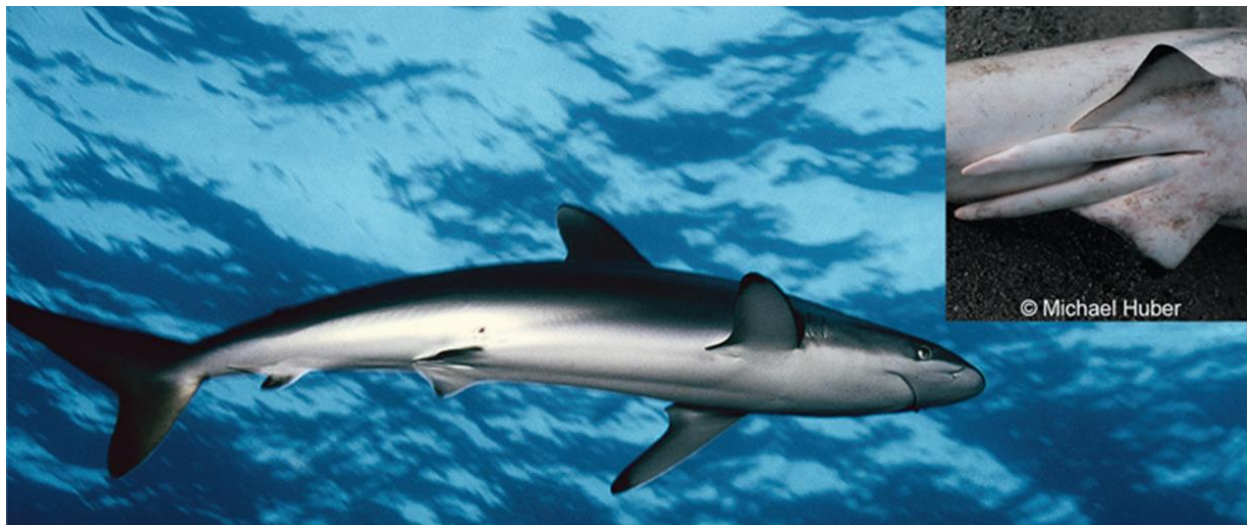
Altered:
Flow Regime & Temperatures

Starved of:
Sediment, Nutrients & Debris
(habitat building blocks)

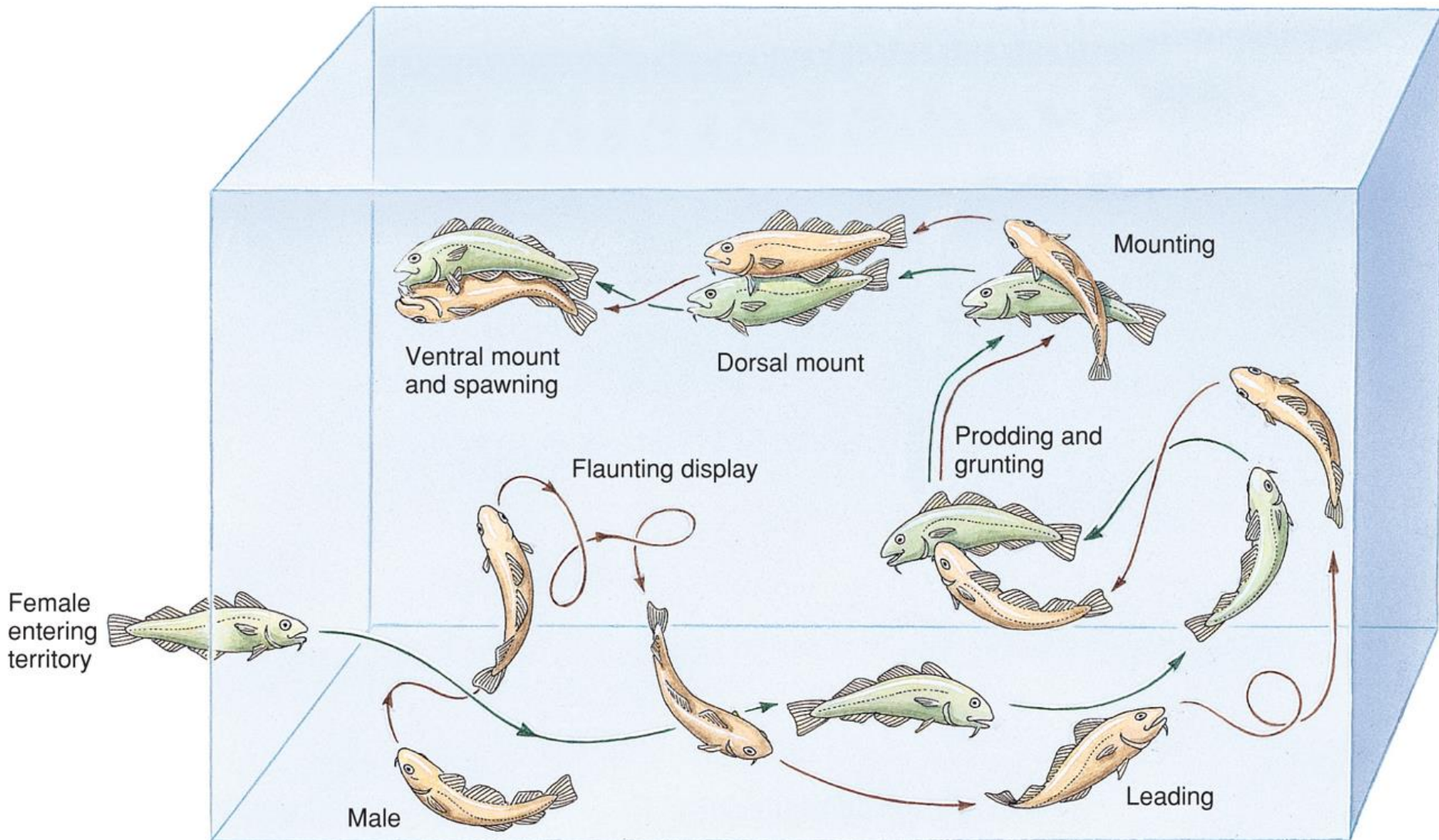


11. Reproduction in Fish

- Sex hormones control the development of sperm and eggs in fish
- The release of sex hormones can be cued by water temperature, day length, specific tide cycles, etc.
- Broadcast spawning is most common
- Some fish do have internal fertilization (sperm is inserted directly into the female by the male)



- Complex mating behaviors are seen in some species





Animal Behaviour
Volume 150, April 2019, Pages 219–237



Behavioural syndromes as a link between ecology and mate choice: a field study in a reef fish population

Current Biology

Volume 14, Issue 13, 13 July 2004, Pages R503–R504



Dispatch

Social Behavior: How Do Fish Find Their Shoal Mate?



Animal Behaviour
Volume 169, November 2020, Pages 119–128



Body-generated hydrodynamic flows influence male–male contests and female mate choice in a freshwater fish



Animal Behaviour
Volume 130, August 2017, Pages 1–7



Male mate choice contributes to behavioural isolation in sexually dimorphic fish with traditional sex roles



Science of The Total Environment
Volume 663, 1 May 2019, Pages 206–215



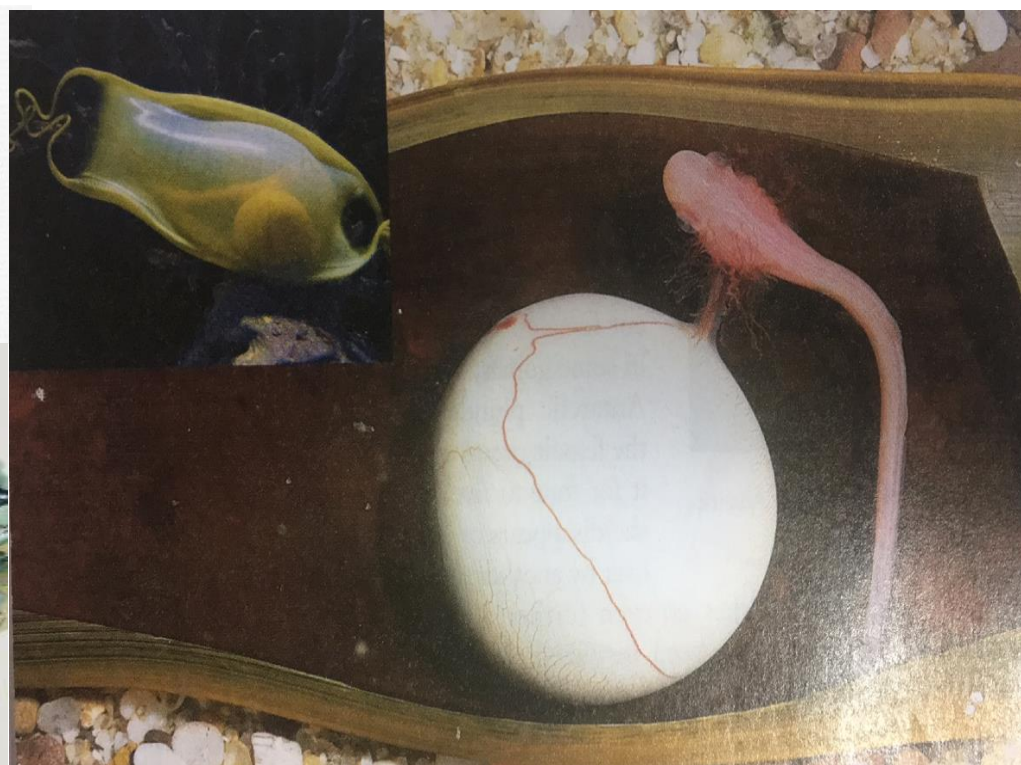
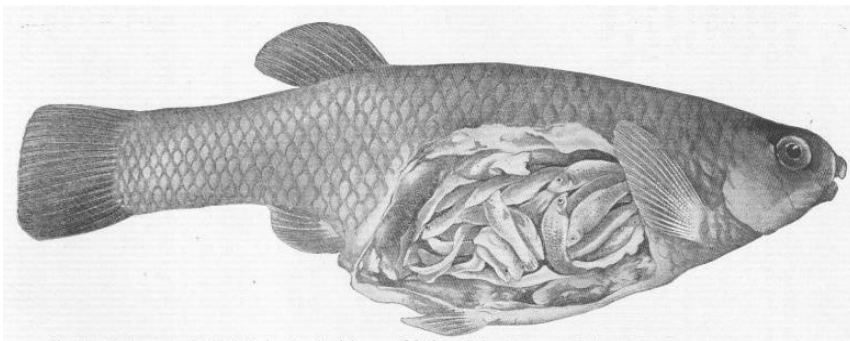
Synthetic progestin etonogestrel negatively affects mating behavior and reproduction in Endler's guppies (*Poecilia wingei*)

- Some fish are hermaphroditic (possess male and female reproductive organs)
 - ▣ Some possess these structures at the same time (simultaneous hermaphrodites), rare, i.e. *Hypoplectrus gummigutta* (Golden hamlets)
 - ▣ Other species possess these structures at different times during the life (sequential hermaphrodites)

Example: clownfish is protandry (雄蕊先熟), fish are first males and develop into females later in life; the cues for these changes are often the result of changes in social structure or environment

Example: Red snapper is protagyny (雌蕊先熟), a large male keeps a “harem” (妻妾群) of females and he fertilizes the eggs of all; When this male dies, the largest female develops into a functional male

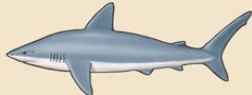
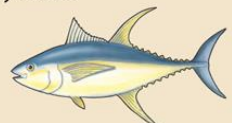
- Depending on the species, fish can either be:
 - ▣ Viviparous – young are born live
 - ▣ Oviparous- egg layers
 - ▣ Ovoviviparous – eggs are kept inside and “hatch” before being released from female reproductive tract



- In most bony fish, eggs are laid by the thousands or millions and are not protected by the parents
- In other bony fish, smaller numbers of eggs are laid and the parent(s) protect the eggs
- **RARE:** A few species are parthenogenic (**单性生殖**), in this reproductive plan, young develop directly from the unfertilized eggs of the female (no DNA from males); the young are “clones” of the female, i.e. zebra shark
- In most fish, intricate behaviors are cues for the release of eggs and sperm (this helps ensure fertilization)
- In some species, color changes or body structure changes may indicate readiness for reproduction

Most Important Characteristics of Marine Fish

Table 8.1 Most Important Characteristics of Marine Fishes

Group	Distinguishing Features	Skeleton	Feeding	Reproduction	Significance in the Marine Environment
Hagfishes 	No paired fins, no scales, exposed gill slits, marine	Cartilaginous skull, no vertebrae, no jaws	Suction by round, muscular mouth with teeth	Oviparous	Predators of dead or dying fishes and bottom invertebrates
Lampreys 	Two dorsal fins only, no scales, exposed gill slits, fresh water or anadromous	Cartilaginous skull, no vertebrae, no jaws	Suction by round, muscular mouth with teeth	Oviparous	Suckers of fish blood or predators of bottom invertebrates
Rays, skates 	Paired fins, placoid scales, pectoral fins greatly expanded, five ventral gill slits, mostly marine	Cartilaginous	Grinding plates to feed on bottom animals or gill rakers to filter plankton	Oviparous, viviparous	Predators of bottom animals or filter feeders
Sharks 	Paired fins, placoid scales, 5–7 exposed and lateral gill slits, mostly marine	Cartilaginous	Teeth in jaws to capture prey or gill rakers to filter plankton	Oviparous, ovoviviparous, viviparous	Predators or filter feeders
Ratfishes 	Paired fins, placoid scales, one pair of gill slits covered by flap of tissue, deep water	Cartilaginous	Grinding plates to feed on bottom invertebrates	Oviparous	Predators of bottom invertebrates
Coelocanth 	Paired fins, large scales, gills covered by operculum, deep water	Bony	Teeth in jaws to capture prey	Ovoviviparous	Predators
Bony fishes 	Paired fins, cycloid or ctenoid scales (absent in some), gills covered by operculum, marine and fresh water	Bony	Teeth in jaws to capture prey or graze or gill rakers to filter plankton	Oviparous, ovoviviparous, viviparous	Predators, grazers, or filter feeders